



OV Shashank
Electrical Engineering
Indian Institute of Technology Bombay
Specialization: Microelectronics and VLSI

14D070021
Dual Degree (B.Tech+M.Tech.)
Male
DOB: 24/06/1997

Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2019	9.81
Intermediate/+2	CBSE	RN Podar School	2014	96.80
Matriculation	CBSE	RN Podar School	2012	10.00

SCHOLASTIC ACHIEVEMENTS

- Currently **ranked 1st** based on CPI among 60+ students in the Electrical Engineering Department, IIT Bombay
- Completed a **Minor** in the **Biosciences and Bioengineering** Department with a Minor CPI of **10**
- Three-time recipient of **Institute Academic Prize** awarded to top two students of each department annually
- Awarded an **AP Grade** for outstanding performance in **6 courses** including RF Chip Design and VLSI Design Lab
- Secured a **perfect 10 SPI** (Semester Performance Index) twice - 5th and 8th semesters at IIT Bombay
- Awarded a **Merit Certificate** for being in top **302** in National Standard Examination in Biology (**NSEB**) 2013
- Secured an All India Rank of **258** in JEE Advanced 2014 among 1.28 Lakh shortlisted candidates

RESEARCH EXPERIENCE

Accelerating HB Analysis for RF Circuits (July '18 - Present)
Guide: Prof. Sachin Patkar, Electrical Engineering, IIT Bombay *MTech Thesis*

- Achieved **~900 GFlops**, **2x** what is obtained by the Intel MKL for Dense LU Decomposition on the **Intel Xeon Phi KNL**.
- Implementation of the **Harmonic Balance** Steady-State Circuit Simulation algorithm for high frequency **RF circuits**
- **Modeling** of the MOSFET RF Circuits and other non-linear elements using charge oriented **Modified Nodal Analysis**
- Optimization **Block LU Decomposition** kernel for Block Sparse Matrices using **structure** and **sparsity aware** tuning
- Investigating opportunities on high-performance computing platforms like **GPUs** and **FPGA-multicore** Hybrid systems

Oscillator Design for Particulate Measurement System (July '18 - Present)
Guide: Prof. Maryam Shojaei Baghini, Electrical Engineering, IIT Bombay *RnD Project*

- Designing **Pierce Oscillator** in **UMC 180nm** technology for MEMS resonator in the Particulate Measurement System
- To achieve a frequency stability of **1 ppm** across **process corners** with **temperature** and **humidity** compensation
- **Layout, Fabrication** and Field **testing** of the Oscillator integrated with an ADC and Data Acquisition and Analysis system
- In collaboration with Prof. Ashwin Seshia, **Cambridge University** for developing and characterizing the MEMS Resonator

HPC Exposure and Exploration: IJEP Internship Program (May '17 - July '17)
Guide: Prof. Reiji Suda, Department of Computer Science, University of Tokyo

- Among the **top 3** students selected from the IITs for the **fully-funded** Summer Internship at **UTokyo** in Computer Science
- Achieved a maximum of **~3.5 TFlops** of performance in **Dense Matrix Multiplication** on Nvidia **GTX 1080 GPU**
- Accelerated **beyond baseline**, **SpMV** for multiple sparse matrix storage formats on a **Many-Core NUMA** Intel Xeon
- Obtained significant performance improvement using **communication-avoiding Stencil** Computation algorithm in **MPI**

Accelerating Model Evaluation Phase of ngSPICE (July '16 - Nov '16)
Guide: Prof. Sachin Patkar, Electrical Engineering, IIT Bombay *Seminar*

- Studied the **Look-Up Table** based approach as an alternate **Model Evaluation algorithm** in the ngSPICE frontend
- Accelerated by **2x** the Tri-Cubic **spline interpolation** based on heuristics and **Horner's Method** of polynomial factorization
- Programmed in **CUDA** for Nvidia GPUs and studied effects of **Register Spills** and **Streaming Processor Occupancy**

COURSE PROJECTS

VCO Design and Layout | Radio Frequency Microelectronics Chip Design (Mar '18 - May '18)

- Awarded the **1st Prize** in the **Layout** Design Competition judged by **industry experts** in a class of **~40** students
- Implemented an LC based **current-reuse Voltage-Controlled Oscillator** architecture with a tuning range of 5 to 6 GHz
- Used capacitor banks and **varactor** for frequency tuning and achieved a **low Phase Noise** of -100 dBc/Hz at 1MHz Offset

Integer Factorization using QFS for Cryptanalysis | VLSI Design Lab (Jan '18 - May '18)

- Studied and implemented the **Quadratic Field Sieve** Algorithm in C++ and an HLS tool: **Bluespec System Verilog**
- Analysed **Sparse Boolean** Matrix structure and coded an optimized **Matrix-Vector** multiplication (BMVM) in Bluespec
- Created a **High-Precision** Integer class in C++ to operate on large numbers **without dynamic memory** allocation
- Accelerated the critical steps of **Block-Wiedemann** Algorithm and **Berlekamp-Massey** Algorithm using hardware loops

PODEM and Deductive Fault Simulation | *Testing and Verification of VLSI Circuits* (Feb '18 - Apr '18)

- Implemented **PODEM** combinational stuck-at fault test-vector generation algorithm in **C++** with **Verilog netlist parsing**
- Designed a backend of **Event-Driven** Logic simulation using **9-Valued** Logic capable of handling **multiple stuck-at** faults
- Used **Deductive Fault Simulation** to minimise the set of test vectors for maximum fault-coverage by using fault dropping

IITB RISC Processor | *Microprocessors and Processor Design* (Feb '18 - Apr '18)

- Designed and Implemented a **16-Bit, 6-Stage Pipelined** RISC processor based on a Turing-Complete ISA in VHDL
- Successfully tested the implementation on a Cyclone IV E FPGA along with a self-designed **UART based DMA bootloader**
- Included **branch predictor**, data and control **hazard mitigation** unit and **result forwarding** unit to improve performance

Operational Transconductance Amp. Design and Layout | *CMOS Analog VLSI Design* (Sep '17 - Nov'17)

- Received **2nd Prize** in the **Layout and Design** Competition among ~90 class students of the graduate course
- Designed a fully differential input, single-ended output 2-Stage **High-Swing Telescopic** Cascode OTA with compensation
- Achieved DC Gain of **95dB**, Phase Margin of **66°**, Power of **0.84 mW** and input-referred Thermal Noise of **12.8nV/ $\sqrt{Hz}$**

Flash A/D Converter Design | *Mixed-Signal VLSI Design* (Mar '18 - Apr '18)

- Designed two **4-Bit 2GS/s** Flash ADC using **Strong-Arm Latch & Class AB** Comparator with reference subtraction
- Designed a **differential T/H** circuit with dummy switches for **clock feedthrough and charge-injection** compensation
- Integrated T/H circuit and performed elaborate **characterization** and **comparison** of the two comparators based ADCs

High-Level Synthesis | *Foundation of VLSI CAD* (Sep '16 - Nov '16)

- Implemented the HLS flow, generating **synthesisable VHDL** Code, of **Scheduling**, and **Binding** for arithmetic expressions
- Optimized hardware synthesis with **shared variable** identification under **resource and timing constraints** input on a GUI

Password Manager | *Data Structures and Algorithms* (Jan '17 - Apr '17)

- Implemented a Password Manager in Visual C++ complete with a GUI, **random password generation** and multiple users
- Fast Searching implemented using **compressed suffix tries**, encryption using **AES** and **SHA2** for master-password hashing

Temperature Controller and IV Characterizer | *Electronic Design Lab* (Jan '17 - Apr '17)

- Architected a system to **automate Current vs Voltage** characterization at various temperatures using a **Peltier Cooler**
- Designed PCBs to measure and **handle 4A** current, using LM35 for temperature measurement and **TivaC** as the controller
- Achieved stable temperature control over a wide range of **-10 to 80°C** with precision of 1°C using a **Bang-Bang** controller

POSITION OF RESPONSIBILITY

Graduate Teaching Assistant | *CMOS Analog IC Design* (July '18 - Present)

- Assisted the professor in managing **logistics**, **course plan** and ensuring smooth functioning of the course
- Responsible for conducting **tutorials**, **designing assignments and projects** and evaluating submissions of ~**140 students**

Junior Design Engineer: Data Acquisition and CAN | *IIT Bombay Racing* (Aug '15 - July '16)

- Implemented **Controller Area Network** (CAN) protocol to improve reliability and reduce wiring harness in a team of 6
- Designed onboard data logging system and real-time **wireless remote** data analysis with a **dashboard GUI** system

Teaching Assistant | *Bridge Courses* (July 2018)

- Devised and conducted an extensive **VHDL and ngSPICE** tutorial for new MTech and PhD entrants of the EE Department
- Helped in **coordinating** and ensuring smooth running of the **lecture** on basics of Electronic Systems and MOSFETs

TECHNICAL SKILLS & INTERESTS

HDLs & Programming	C++, C, Python, MATLAB, VHDL, Bluespec System Verilog, CUDA, MPI, Assembly
Design Tools	Cadence Virtuoso, Intel Quartus, Xilinx Vivado, SDX and SDSoc, TI CCS
Interests	Circuit Simulation, RF and Mixed Signal Design, Processor Design, and Digital Design

RELEVANT COURSES

Digital VLSI	VLSI Design, Testing and Verification of VLSI , Foundation of VLSI CAD , VLSI Design Lab
Analog VLSI	CMOS Analog VLSI Design, Mixed-Signal VLSI Design, RF Microelectronics Chip Design
Processor Design	Advanced Topics in Computer Architecture, Advanced Processor Design, Microprocessors

EXTRA CURRICULAR ACTIVITIES

- Volunteered in the NSS **Summer Volunteering** Program by teaching at Computer Literacy Program May - Jun '15
- Volunteered in a $\text{\LaTeX}$ workshop jointly conducted by the PG Academic Council and IRSCP Dec '17
- Learning the **Japanese** Language: Completed 100 hours with A+ grade Aug '17 - Present
- Recorded magazines for the **Lend Your Voice** initiative through the National Service Scheme, IIT Bombay 2014 - 15